EXPERIMENT NO. 1 B (Group A)

➢ Aim: Apply LDA Algorithm on Iris Dataset and classify which species a given flower belongs to Dataset Link:https://www.kaggle.com/datasets/uciml/iris

Name:Aditya Rade Roll.No:BEAD23167

Outputs:

Step 1 — Import and Load Iris Dataset

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("Iris.csv")

print("Dataset loaded successfully!")
print("Shape:", df.shape)

display(df.head())
print("\nColumns:")
print(df.columns.tolist())
```

```
Dataset loaded successfully!
Shape: (150, 6)
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
Columns:
['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm', 'Species']
```

Step 2 — Prepare the Data

In [2]:
```
from sklearn.preprocessing import StandardScaler

# Separate input features and target
X = df.drop(["Id", "Species"], axis=1)
y = df["Species"]

print("Species:")
print(y.unique())

# Scale the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
print("\nFeatures scaled successfully!")
print("Feature shape:", X_scaled.shape)
```

```
Species:
['Iris-setosa' 'Iris-versicolor' 'Iris-virginica']

Features scaled successfully!
Feature shape: (150, 4)
```

Step 3 — Apply LDA

In [3]:
```
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis

# Create LDA model
lda = LinearDiscriminantAnalysis(n_components=2)

# Transform data using LDA
X_lda = lda.fit_transform(X_scaled, y)

print("LDA applied successfully!")
print("LDA shape:", X_lda.shape)
```

```
LDA applied successfully!
LDA shape: (150, 2)
```

Step 4 — Train Classification Model

In [4]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X_lda,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Train classifier
classifier = LogisticRegression()
classifier.fit(X_train, y_train)

# Test prediction
y_pred = classifier.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Classification completed!")
print("Accuracy:", accuracy)
```

```
Classification completed!
Accuracy: 1.0
```

Step 5 — Predict a New Flower

In [5]:
```
# New flower measurements
new_flower = [[6.7, 3.0, 5.2, 2.3]]

# Scale using the same scaler
new_flower_scaled = scaler.transform(new_flower)

# Apply the same LDA transformation
new_flower_lda = lda.transform(new_flower_scaled)

# Predict species
predicted_species = classifier.predict(new_flower_lda)

print("Predicted species:", predicted_species[0])
```

```
Predicted species: Iris-virginica
```

```
c:\Users\adity\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but StandardScaler was fitted with feature names
  warnings.warn(
```

In [7]:
```
new_flower = pd.DataFrame(
    [[6.7, 3.0, 5.2, 2.3]],
    columns=X.columns
)

new_flower_scaled = scaler.transform(new_flower)
new_flower_lda = lda.transform(new_flower_scaled)

predicted_species = classifier.predict(new_flower_lda)

print("Predicted species:", predicted_species[0])
print("Name: Aditya Rade  Roll.No:BEAD23167")
```

```
Predicted species: Iris-virginica
Name: Aditya Rade  Roll.No:BEAD23167
```